

Case study - iviRA model

Xavier G.L.V. Pouwels

Introduction

This vignette shows how to use the functionalities of `pacheck` on the `iviRA` health economic model.

Loading packages and data

```
#library(pacheck)
devtools::load_all()
```

i Loading `pacheck`

Warning: replacing previous import 'glmnet::na.replace' by 'gtools::na.replace' when loading 'pacheck'

Warning: replacing previous import 'boot::logit' by 'gtools::logit' when loading 'pacheck'

Warning: replacing previous import 'boot::inv.logit' by 'gtools::inv.logit' when loading 'pacheck'

Warning: replacing previous import 'dplyr::lag' by 'stats::lag' when loading 'pacheck'

Warning: replacing previous import 'dplyr::filter' by 'stats::filter' when loading 'pacheck'

Warning: replacing previous import 'magrittr::is_less_than' by 'testthat::is_less_than' when loading 'pacheck'

Warning: replacing previous import 'magrittr::not' by 'testthat::not' when loading 'pacheck'

Warning: replacing previous import 'magrittr::equals' by 'testthat::equals' when loading 'pacheck'

Warning: replacing previous import 'dplyr::matches' by 'testthat::matches' when loading 'pacheck'

Warning: replacing previous import 'assertthat::has_name' by 'tibble::has_name' when loading 'pacheck'

Warning: replacing previous import 'magrittr::extract' by 'tidyr::extract' when loading 'pacheck'

Warning: replacing previous import 'testthat::matches' by 'tidyr::matches' when loading 'pacheck'

```
data("l_iviRA_pa_params")
data("l_iviRA_out_summ_1")
data("l_iviRA_out_summ_2")
```

Generating the probabilistic inputs and outputs

The source code to generate the model inputs and outputs of the `iviRA` cost-effectiveness model is available in the source code of the `pacheck` package on github (https://github.com/Xa4P/pacheck/blob/master/analysis/03-iviRA_data_generation.R).

Preparing dataset for use in pacheck

The following code chunk prepares the inputs and outputs list of the `iviRA` package to use the `pacheck` tests. In this example, only a subset of all inputs is used for convenience.

```

df_iviRA_pa <- data.frame(
  p_discount = l_iviRA_pa_params$tx.cost$discount,
  p_rebound = l_iviRA_pa_params$rebound,
  or_mort = exp(l_iviRA_pa_params$mort.logor),
  r_haq_prog = l_iviRA_pa_params$haq.lprog.tx,
  n_hosp_days = l_iviRA_pa_params$hosp.cost$hosp.days,
  c_hosp_pday = l_iviRA_pa_params$hosp.cost$cost.pday,
  c_si = l_iviRA_pa_params$si.cost,
  c_prod_loss = l_iviRA_pa_params$prod.loss,
  u_model_wailoo = l_iviRA_pa_params$utility.wailoo,
  t_qaly_thx1 = l_iviRA_out_summ_1$means$qalys,
  t_qaly_d_thx1 = l_iviRA_out_summ_1$means$dqalys,
  t_cost_d_thx1 = l_iviRA_out_summ_1$means$dtot_cost,
  t_qaly_thx2 = l_iviRA_out_summ_2$means$qalys,
  t_qaly_d_thx2 = l_iviRA_out_summ_2$means$dqalys,
  t_cost_d_thx2 = l_iviRA_out_summ_2$means$dtot_cost
)
df_iviRA_pa$inc_qaly <- df_iviRA_pa$t_qaly_d_thx1 - df_iviRA_pa$t_qaly_d_thx2
df_iviRA_pa$inc_cost <- df_iviRA_pa$t_cost_d_thx1 - df_iviRA_pa$t_cost_d_thx2

```

Testing plausibility of model inputs and outputs

One way to test the plausibility of model inputs and outputs is simply to inspect the summary statistics of each variable. This can be done with the `pacheck` package using the `generate_sum_stats` function

```

# Summary stats
generate_sum_stats(df = df_iviRA_pa)

```

	Parameter	Mean	SD	Percentile_2.5th
1	p_discount.abtiv	0.249	0.028	0.202
2	p_discount.abtsc	0.250	0.030	0.202
3	p_discount.ada	0.248	0.029	0.202
4	p_discount.adabiosbwwd	0.249	0.029	0.203
5	p_discount.ana	0.251	0.029	0.203
6	p_discount.bct	0.250	0.029	0.202
7	p_discount.czp	0.250	0.029	0.203
8	p_discount.etn	0.250	0.030	0.201
9	p_discount.etnbiosszzs	0.250	0.029	0.202
10	p_discount.etnbiosykro	0.250	0.029	0.203

11	p_discount.gol	0.249	0.029	0.202
12	p_discount.hcl	0.250	0.029	0.203
13	p_discount.ifx	0.250	0.028	0.204
14	p_discount.ifxbiosqbtx	0.251	0.029	0.204
15	p_discount.cdmards	0.249	0.029	0.204
16	p_discount.rtx	0.250	0.029	0.202
17	p_discount.sar	0.250	0.029	0.202
18	p_discount.ssz	0.247	0.029	0.202
19	p_discount.tcz	0.250	0.029	0.203
20	p_discount.tof	0.250	0.029	0.203
21	p_discount.upa	0.250	0.029	0.202
22	p_rebound	0.847	0.086	0.708
23	or_mort	2.238	0.238	1.808
24	r_haq_prog.cdmards	0.031	0.003	0.026
25	r_haq_prog.abtivismtx	-0.001	0.002	-0.004
26	r_haq_prog.abtscmtx	-0.001	0.001	-0.004
27	r_haq_prog.adamtx	-0.003	0.007	-0.018
28	r_haq_prog.ada	0.012	0.007	-0.003
29	r_haq_prog.adabiosbwwdmtx	-0.003	0.008	-0.018
30	r_haq_prog.anamtx	-0.001	0.001	-0.004
31	r_haq_prog.bct	-0.001	0.001	-0.004
32	r_haq_prog.bctmtx	-0.001	0.001	-0.004
33	r_haq_prog.czp	-0.001	0.002	-0.004
34	r_haq_prog.czpmtx	-0.001	0.002	-0.004
35	r_haq_prog.etn	-0.003	0.004	-0.011
36	r_haq_prog.etnmtx	-0.005	0.003	-0.012
37	r_haq_prog.etnbioszzsmtx	-0.005	0.003	-0.011
38	r_haq_prog.etnbiosykromtx	-0.005	0.003	-0.012
39	r_haq_prog.gol	-0.001	0.001	-0.004
40	r_haq_prog.golmtx	-0.001	0.001	-0.004
41	r_haq_prog.ifxmtx	-0.001	0.004	-0.009
42	r_haq_prog.ifxbiosqbtxmtx	-0.001	0.004	-0.008
43	r_haq_prog.placebo	0.031	0.003	0.026
44	r_haq_prog.nbt	0.031	0.003	0.026
45	r_haq_prog.rtx	-0.001	0.002	-0.004
46	r_haq_prog.rtxmtx	-0.001	0.002	-0.004
47	r_haq_prog.sar	-0.001	0.002	-0.004
48	r_haq_prog.sarmtx	-0.001	0.002	-0.004
49	r_haq_prog.triple	-0.001	0.002	-0.004
50	r_haq_prog.tcz	-0.001	0.002	-0.004
51	r_haq_prog.tczmtx	-0.001	0.002	-0.004
52	r_haq_prog.tofmtx	-0.001	0.002	-0.004
53	r_haq_prog.tof	-0.001	0.002	-0.004

54	r_haq_prog.upamtx	-0.001	0.002	-0.004		
55	n_hosp_days.haq_less0.5	0.250	0.494	0.000		
56	n_hosp_days.haq0.5to1	0.113	0.469	0.000		
57	n_hosp_days.haq1to1.5	0.513	0.498	0.015		
58	n_hosp_days.haq1.5to2	0.711	0.485	0.112		
59	n_hosp_days.haq2to2.5	1.861	0.493	0.995		
60	n_hosp_days.haq2.5plus	4.172	0.516	3.246		
61	c_hosp_pday.haq_less0.5	1347.829	204.953	968.022		
62	c_hosp_pday.haq0.5to1	1346.712	207.486	968.348		
63	c_hosp_pday.haq1to1.5	1348.222	207.233	998.871		
64	c_hosp_pday.haq1.5to2	1332.318	203.471	966.891		
65	c_hosp_pday.haq2to2.5	1352.780	206.132	975.975		
66	c_hosp_pday.haq2.5plus	1353.003	202.203	1000.994		
67	c_si	5881.020	670.976	4749.861		
68	c_prod_loss	6138.815	1617.716	3034.525		
69	u_model_wailoo.int	2.073	0.026	2.023		
70	u_model_wailoo.age	0.006	0.000	0.005		
71	u_model_wailoo.dis_dur	0.002	0.000	0.001		
72	u_model_wailoo.haq0	-0.200	0.010	-0.220		
73	u_model_wailoo.male	-0.291	0.012	-0.316		
74	u_model_wailoo.prev_dmards	0.025	0.003	0.020		
75	u_model_wailoo.haq	-0.865	0.010	-0.884		
76	t_qaly_thx1	14.357	0.999	12.521		
77	t_qaly_d_thx1	10.003	0.560	8.911		
78	t_cost_d_thx1	465826.808	43090.450	381059.903		
79	t_qaly_thx2	12.028	0.819	10.508		
80	t_qaly_d_thx2	8.588	0.470	7.685		
81	t_cost_d_thx2	155392.483	37547.005	86926.983		
82	inc_qaly	1.415	0.513	0.458		
83	inc_cost	310434.325	29391.236	256693.134		
	Percentile_97.5th	Minimum	Maximum	Median	Skewness	Kurtosis
1	0.297	0.200	0.300	0.249	0.044	1.843
2	0.298	0.200	0.300	0.251	-0.029	1.745
3	0.297	0.200	0.300	0.248	0.092	1.843
4	0.297	0.200	0.300	0.248	0.026	1.857
5	0.297	0.200	0.300	0.251	-0.042	1.786
6	0.298	0.200	0.300	0.250	0.035	1.821
7	0.298	0.200	0.300	0.250	0.026	1.732
8	0.298	0.200	0.300	0.250	-0.028	1.786
9	0.297	0.200	0.300	0.250	0.006	1.790
10	0.297	0.200	0.300	0.251	-0.035	1.798
11	0.297	0.200	0.300	0.248	0.038	1.821
12	0.298	0.200	0.300	0.252	-0.058	1.822

13	0.297	0.200	0.300	0.249	0.065	1.841
14	0.297	0.200	0.300	0.250	0.018	1.733
15	0.298	0.200	0.300	0.250	0.040	1.768
16	0.296	0.200	0.300	0.250	-0.040	1.754
17	0.298	0.200	0.300	0.251	-0.027	1.762
18	0.297	0.200	0.300	0.246	0.110	1.822
19	0.297	0.200	0.300	0.250	0.025	1.827
20	0.298	0.200	0.300	0.251	0.000	1.769
21	0.297	0.200	0.300	0.249	-0.017	1.794
22	0.994	0.700	1.000	0.846	0.055	1.835
23	2.747	1.515	2.985	2.238	0.220	2.919
24	0.036	0.022	0.038	0.031	-0.003	2.866
25	0.002	-0.005	0.004	-0.001	0.125	2.862
26	0.002	-0.006	0.004	-0.001	-0.043	3.034
27	0.011	-0.026	0.023	-0.003	0.028	2.881
28	0.026	-0.013	0.033	0.012	-0.125	3.155
29	0.011	-0.034	0.022	-0.003	-0.159	3.132
30	0.002	-0.006	0.004	-0.001	-0.023	2.753
31	0.002	-0.005	0.004	-0.001	-0.053	2.902
32	0.002	-0.005	0.004	-0.001	-0.026	2.878
33	0.002	-0.005	0.004	-0.001	0.032	2.844
34	0.002	-0.006	0.004	-0.001	-0.006	3.005
35	0.004	-0.014	0.006	-0.003	-0.133	2.739
36	0.002	-0.015	0.005	-0.005	-0.076	2.872
37	0.001	-0.016	0.007	-0.005	0.070	2.853
38	0.002	-0.015	0.005	-0.005	-0.080	2.836
39	0.002	-0.006	0.004	-0.001	0.042	2.861
40	0.002	-0.005	0.003	-0.001	-0.001	2.775
41	0.006	-0.014	0.012	-0.001	-0.009	3.012
42	0.006	-0.013	0.012	-0.001	-0.045	3.169
43	0.036	0.024	0.039	0.031	0.029	3.004
44	0.036	0.023	0.039	0.031	-0.059	2.964
45	0.002	-0.006	0.004	-0.001	0.010	2.887
46	0.002	-0.007	0.003	-0.001	-0.128	2.916
47	0.002	-0.006	0.003	-0.001	-0.137	2.939
48	0.002	-0.006	0.004	-0.001	-0.085	3.005
49	0.002	-0.006	0.004	-0.001	0.051	2.971
50	0.002	-0.005	0.004	-0.001	0.078	2.848
51	0.002	-0.006	0.003	-0.001	0.021	2.728
52	0.002	-0.007	0.004	-0.001	0.097	2.891
53	0.002	-0.006	0.004	-0.001	-0.051	2.788
54	0.002	-0.006	0.004	-0.001	-0.051	3.165
55	1.718	0.000	4.988	0.046	3.968	24.943

56	1.119	0.000	6.437	0.000	8.309	90.684
57	1.860	0.002	3.546	0.349	1.782	7.034
58	1.955	0.044	3.552	0.600	1.400	5.640
59	2.929	0.535	4.276	1.827	0.517	3.734
60	5.235	2.788	6.715	4.144	0.351	3.485
61	1753.556	753.079	1990.579	1343.707	0.161	2.795
62	1801.829	784.276	2198.674	1338.737	0.353	3.388
63	1786.713	749.125	2128.686	1335.141	0.384	3.110
64	1743.301	807.784	2117.726	1316.352	0.341	3.127
65	1781.199	765.837	2191.253	1334.969	0.330	3.249
66	1780.244	849.388	2045.042	1346.596	0.313	2.818
67	7003.707	4707.990	7047.494	5899.240	-0.065	1.834
68	9365.375	888.422	11144.503	6152.839	-0.030	2.869
69	2.122	1.984	2.154	2.073	0.001	2.630
70	0.007	0.005	0.007	0.006	-0.004	3.188
71	0.003	0.001	0.004	0.002	-0.107	3.263
72	-0.179	-0.232	-0.170	-0.200	-0.003	2.938
73	-0.267	-0.333	-0.248	-0.291	-0.103	3.184
74	0.030	0.016	0.034	0.025	0.051	3.134
75	-0.844	-0.901	-0.838	-0.865	0.039	2.979
76	16.341	10.991	17.989	14.360	0.159	3.145
77	11.139	7.973	11.778	10.005	0.082	3.068
78	550228.215	339915.846	581134.804	464629.195	0.043	2.758
79	13.660	9.240	16.013	12.025	0.110	3.409
80	9.510	6.894	10.785	8.584	0.046	3.348
81	232320.584	43118.720	279234.477	154934.153	0.128	3.017
82	2.401	-0.269	2.892	1.391	0.053	2.739
83	366739.132	223195.938	413151.466	309897.706	0.109	2.907

```
# Specific checks
```

```
## Proportions & probabilities between 0 and 1
```

```
v_vars_discount <- grep("p_discount", names(df_ivIRA_pa), value = TRUE) # returns variable names
```

```
check_binary(c(v_vars_discount, "p_rebound"), df = df_ivIRA_pa)
```

	Input	Negative_values	Values_above_1
1	p_discount.abtiv	None	None
2	p_discount.abtsc	None	None
3	p_discount.ada	None	None
4	p_discount.adabiosbwtd	None	None
5	p_discount.ana	None	None
6	p_discount.bct	None	None
7	p_discount.czp	None	None

8	p_discount.etn	None	None
9	p_discount.etnbioszzs	None	None
10	p_discount.etnbiosykro	None	None
11	p_discount.gol	None	None
12	p_discount.hcl	None	None
13	p_discount.ifx	None	None
14	p_discount.ifxbiosqbtx	None	None
15	p_discount.cdmards	None	None
16	p_discount.rtx	None	None
17	p_discount.sar	None	None
18	p_discount.ssz	None	None
19	p_discount.tcz	None	None
20	p_discount.tof	None	None
21	p_discount.upa	None	None
22	p_rebound	None	None

```
## Costs, counts, OR positive
v_vars_n <- grep("n_", names(df_ivIRA_pa), value = TRUE)
v_vars_c <- grep("c_", names(df_ivIRA_pa), value = TRUE)
v_vars_c <- v_vars_c[-grep("inc_", v_vars_c)]
check_positive(c(v_vars_n, v_vars_c, "or_mort"), df = df_ivIRA_pa)
```

	Input	Negative_values
1	n_hosp_days.haq_less0.5	None
2	n_hosp_days.haq0.5to1	None
3	n_hosp_days.haq1to1.5	None
4	n_hosp_days.haq1.5to2	None
5	n_hosp_days.haq2to2.5	None
6	n_hosp_days.haq2.5plus	None
7	c_hosp_pday.haq_less0.5	None
8	c_hosp_pday.haq0.5to1	None
9	c_hosp_pday.haq1to1.5	None
10	c_hosp_pday.haq1.5to2	None
11	c_hosp_pday.haq2to2.5	None
12	c_hosp_pday.haq2.5plus	None
13	c_si	None
14	c_prod_loss	None
15	or_mort	None

```
do_check(df = df_ivIRA_pa,
         v_vars = c(v_vars_n, v_vars_c, "or_mort"),
```



```
check = ~ .x >= 0,
label_check = "positive") # alternative
```

```
$check
n_hosp_days.haq_less0.5  n_hosp_days.haq0.5to1  n_hosp_days.haq1to1.5
                        TRUE                        TRUE                        TRUE
  n_hosp_days.haq1.5to2  n_hosp_days.haq2to2.5  n_hosp_days.haq2.5plus
                        TRUE                        TRUE                        TRUE
c_hosp_pday.haq_less0.5  c_hosp_pday.haq0.5to1  c_hosp_pday.haq1to1.5
                        TRUE                        TRUE                        TRUE
  c_hosp_pday.haq1.5to2  c_hosp_pday.haq2to2.5  c_hosp_pday.haq2.5plus
                        TRUE                        TRUE                        TRUE
                        c_si                    c_prod_loss                or_mort
                        TRUE                        TRUE                        TRUE
```

```
$messages
# A tibble: 1 x 2
  ok      message
<lgl> <glue>
1 TRUE  all variables are positive
```

```
## Discounted results lower than undiscounted results
do_discount_check(df = df_iviRA_pa,
  v_outcomes = c("t_qaly_thx1", "t_qaly_thx2"),
  v_outcomes_d = c("t_qaly_d_thx1", "t_qaly_d_thx2"))
```

Test passed

	Test Result
1 All discounted outcomes are lower than undiscounted outcomes	TRUE

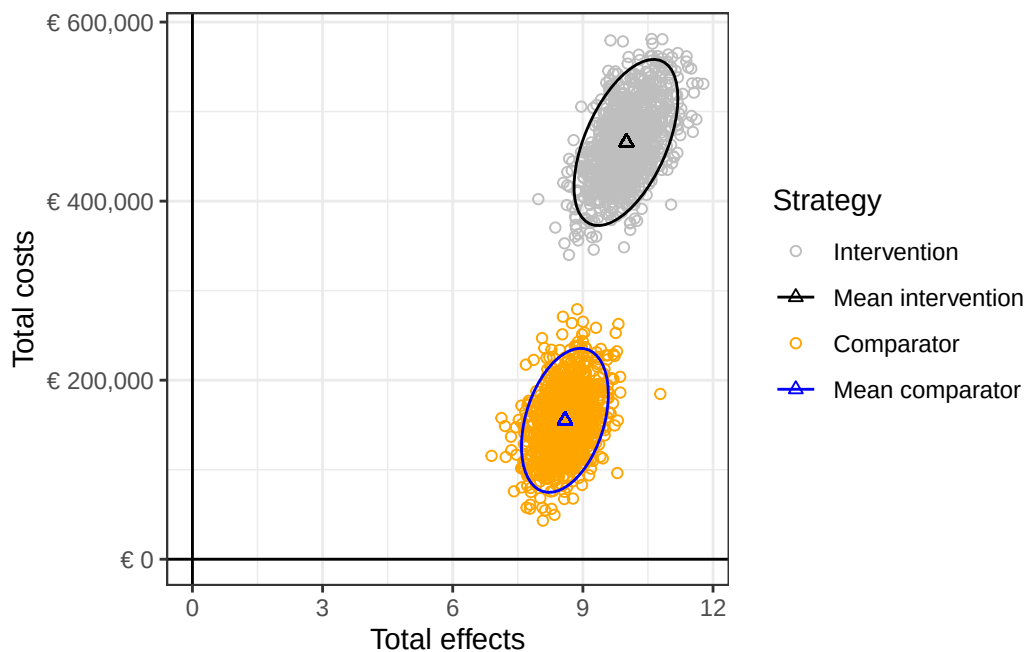
Drawing health economic figures

pacheck also contains figures to draw standard figures for health economic evaluations, such as the cost-effectiveness plane (`plot_ce` function), the incremental cost-effectiveness plane (`plot_ice` function), the cost-effectiveness acceptability curves (`plot_ceac` function), and net benefit curves (`plot_nb` function). Both the `plot_ceac` and `plot_nb` functions require data frames containing specific variable names, which can respectively be obtained with the accompanying functions `calculate_ceac` and `calculate_nb`.

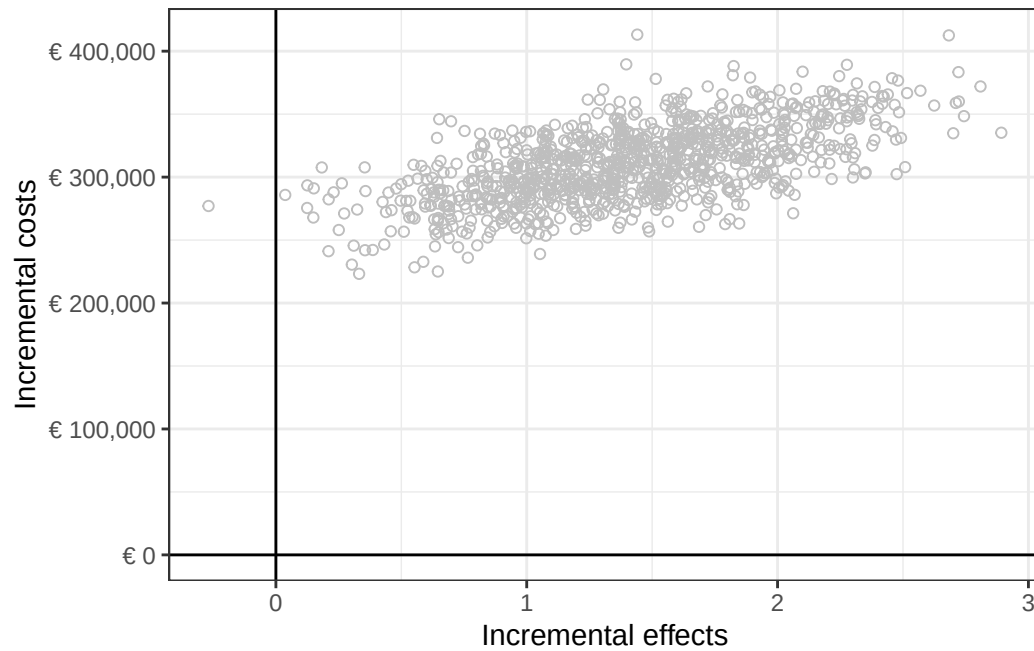
```
plot_ce(df = df_iviRA_pa,
        e_int = "t_qaly_d_thx1",
        e_comp = "t_qaly_d_thx2",
        c_int = "t_cost_d_thx1",
        c_comp = "t_cost_d_thx2")
```

Warning in ggplot2::geom_point(data = df, ggplot2::aes_string(x = mean(df[, : All aesthetics
i Please consider using `annotate()` or provide this layer with data containing
a single row.

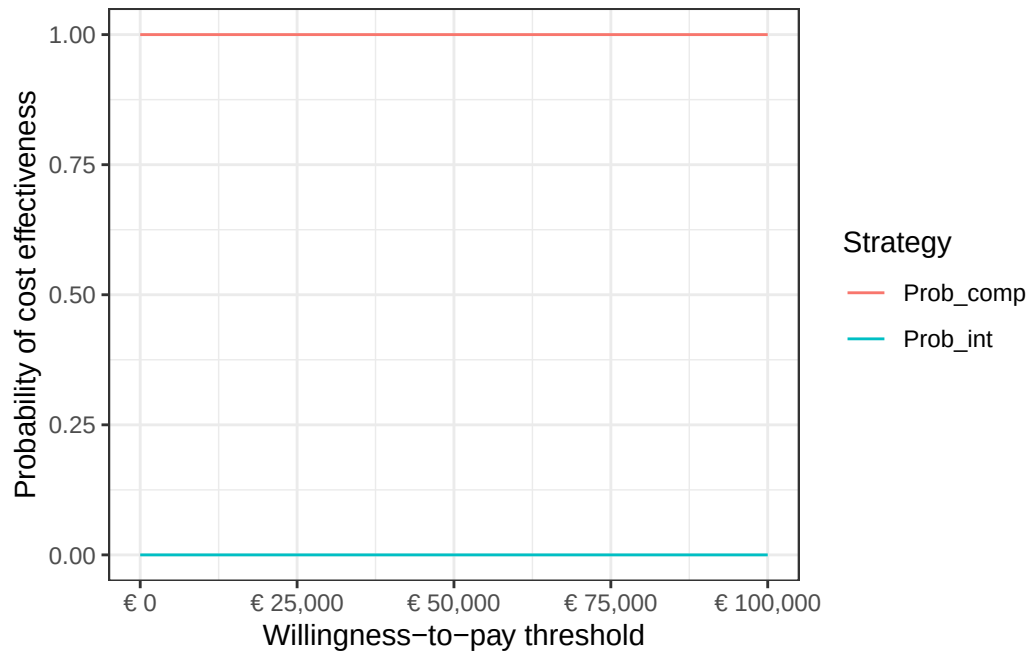
Warning in ggplot2::geom_point(data = df, ggplot2::aes_string(x = mean(df[, : All aesthetics
i Please consider using `annotate()` or provide this layer with data containing
a single row.



```
plot_ice(df = df_iviRA_pa,
         e_int = "t_qaly_d_thx1",
         e_comp = "t_qaly_d_thx2",
         c_int = "t_cost_d_thx1",
         c_comp = "t_cost_d_thx2")
```



```
df_ceac <- calculate_ceac(df = df_iviRA_pa,  
  e_int = "t_qaly_d_thx1",  
  e_comp = "t_qaly_d_thx2",  
  c_int = "t_cost_d_thx1",  
  c_comp = "t_cost_d_thx2")  
plot_ceac(df = df_ceac,  
  name_wtp = "WTP_threshold")
```



```
df_nb <- calculate_nb(df = df_iviRA_pa,  
  e_int = "t_qaly_d_thx1",  
  e_comp = "t_qaly_d_thx2",  
  c_int = "t_cost_d_thx1",  
  c_comp = "t_cost_d_thx2",  
  wtp = 80000)  
plot_nb(df = df_nb)
```

